Chaithanya Vemuri

Junior Data Engineer / ML Engineer | Python, SQL & Reliable ML Systems

Berlin, Germany | chaithanyavemuri28@gmail.com
github.com/nikchey29 | Portfolio & case studies | LinkedIn

PROFILE

Berlin-based MSc candidate building data pipelines, recommendation systems and forecasting experiments. My independent work connects data validation and chronological evaluation to API serving and delivery labs. Seeking junior Data Engineering or ML Engineering opportunities.

TECHNICAL SKILLS

Data: Python, SQL, DuckDB, PostgreSQL, Parquet, dbt, pandas; PySpark / Redpanda project work

ML / services: scikit-learn, PyTorch, MLflow, chronological evaluation, FastAPI, Docker, pytest

Delivery labs: GitHub Actions, Terraform, GCP/GKE, Helm, Argo CD, Prometheus / Grafana

SELECTED INDEPENDENT ENGINEERING & MSc RESEARCH

REES46 V2 / Data engineering / recommendations

- Processed 411.7M historical events into 410.3M canonical Silver events using DuckDB and Parquet. Reworked a failed ~30.6 GiB global spill into restartable monthly Gold outputs and 64 session hash buckets.
- Separate offline benchmark: 1,499,021 training interactions and 5,000 held-out test users. Purchase-ranker MRR@20 improved from 0.02443 to 0.03556 versus popularity; exposed the bundle through a local FastAPI service.

OTTO hourly cart / order forecasting / MSc dissertation

- Compared eight methods on 672 hourly observations using three chronological development folds and a separate 96-hour holdout. Weekly seasonal baseline selected at 13.93% mean CV WAPE; final carts / orders WAPE: 9.80% / 11.34%. The Transformer was not selected.

FulfillAI / Synthetic data / ML / API platform

- Built a workflow using 50,000 synthetic orders with PostgreSQL/dbt, time-aware features, streaming and API serving. Frozen demand WAPE was 69.59% versus 88.24% for rolling-28; RMSE worsened from 0.836723 to 0.933770.
- Extended the API into a controlled Terraform / GCP / GKE lab with keyless CI, exact-digest GitOps promotion, Helm / Argo CD, Prometheus / Grafana and documented recovery exercises.

EDUCATION & TRAINING

MSc Data Science, AI & Digital Business | Gisma University of Applied Sciences | In progress

BTech Electronics & Communication Engineering | Sri Venkateswara College of Engineering, Tirupati

IntrnForte: Python self-directed learning / training.